Genomic Recurrence-Risk Service (70-gene profile)

275 Riverside Parkway, Edgewater, NY 10010
CLIA 33D7711999
Service Director: Ito, Renata K., PhD, Director

70-GENE RECURRENCE RISK PROFILE — INVASIVE BREAST CARCINOMA — FINAL REPORT

PATIENT	SPECIMEN	ORDERING
BERGSTROM, ANITA L. MRN MRN-5519082 52 y / Female	Lab accession GRRS-26-RR-00731 Client WCI-26-S-04115 FFPE A4 Received 2026-05-02	Westhaven Cancer Institute T. Okafor, MD Reported 2026-05-09

RECURRENCE RISK RESULT

BINARY CALL: **LOW RISK**
Continuous index: 0.18 (cutoff >0.31 = High Risk)
10-year distant-recurrence risk (estimated): approximately 8% without adjuvant chemotherapy
Sensitivity for chemotherapy benefit: Low — minimal predicted absolute benefit from adjuvant chemotherapy in this Low Risk genomic category

CLINICAL CONTEXT

This 70-gene expression profile is a high-complexity laboratory developed test for early-stage hormone-receptor-positive HER2-negative invasive breast carcinoma. The binary Low Risk / High Risk call combined with classical clinicopathologic features (tumor size, grade, nodal status, age) is intended to inform the decision to add adjuvant chemotherapy to endocrine therapy. In this case the patient's clinical-pathologic risk is intermediate (2.1 cm, grade 2, ER+/PR+/HER2-, pN0, postmenopausal). A Low Risk genomic call in an intermediate clinical-risk setting supports an endocrine-therapy-only adjuvant approach in most patients, after shared decision-making.

METHODOLOGY

Total RNA was extracted from FFPE block A4 and quantified by fluorometric assay. Reverse transcription and amplification performed on a 70-gene custom NanoString-equivalent codeset. Normalized expression values were submitted to the laboratory's validated 70-gene index model, which returns a continuous recurrence-risk index (0.00 — 1.00) with a Bayesian binary low/high cutoff at 0.31. The cutoff is anchored to the original MINDACT-style 10-year distant-recurrence outcome cohort. Analytical performance: technical reproducibility CV < 0.04 on internal controls. This is a high-complexity laboratory-developed test; performance characteristics were established by Genomic Recurrence-Risk Service (70-gene profile). Not FDA-cleared.

LIMITATIONS

The recurrence-risk profile is designed for hormone-receptor-positive, HER2-negative, lymph-node-negative or limited node-positive (≤3 positive nodes) early invasive breast carcinoma. The result should be interpreted alongside clinical-pathologic risk and patient preferences. The test does not address response to specific endocrine agents and does not predict CDK4/6 inhibitor benefit. Tumor cellularity in the submitted block was estimated at 55%; acceptable but on the lower end of the technical range.

Electronically signed: Ito, Renata K., PhD, Director, Service Director, and Sandhu, J., PhD, Bioinformatics Lead — 2026-05-09.